

Assignment 1

Consider a potential well

$$V(x) = \begin{cases} 0 & x < 0 \\ -V_0 & 0 < x < a \\ 0 & x > 0 \end{cases}$$

Let $E = -f$, $f > 0$, be the the bound state energy eigenvalue of the Schroedinger equation for this this potential well. It can be showed that it satisfies the equation

$$\tan(za) = \frac{2\sqrt{(z_0/z)^2 - 1}}{2 - (z_0/z)^2} \quad (1)$$

where $z_0^2 \equiv 2mV_0/\hbar^2$, $z^2 = z_0^2 - 2mf/\hbar^2$. Take $a = 1$, $V_0 = 1$, $m = 0.5$, $\hbar = 1$. $z = 0$ is a trivial solution of this equation. Calculate a non-trivial energy eigenvalue with the Newton method. Your program should output the energy eigenvalue accurate to 7 decimal places.